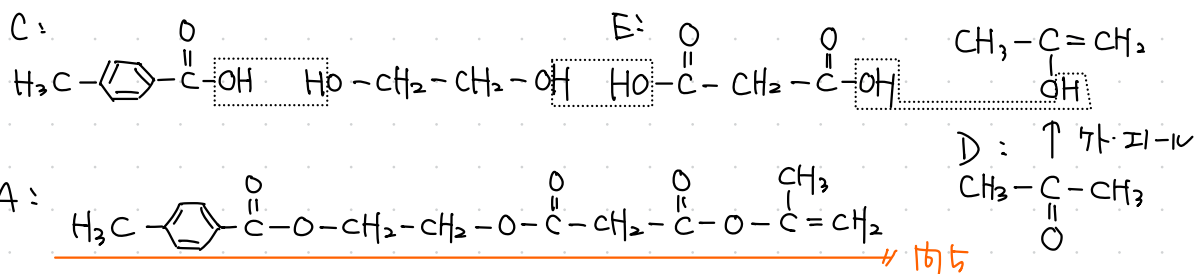
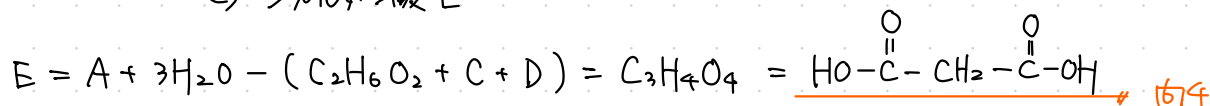
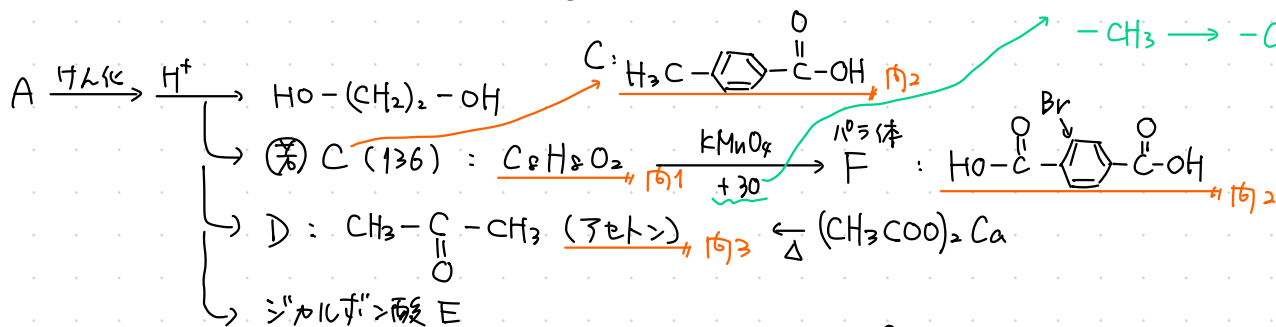
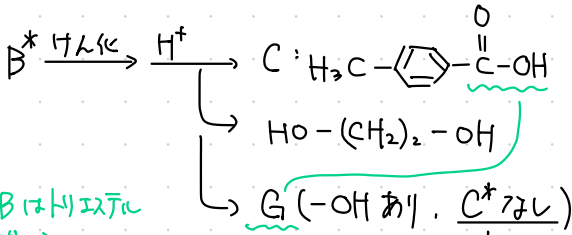
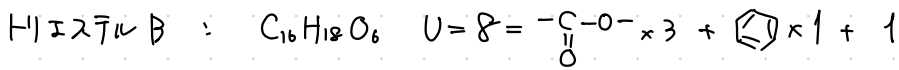


1 THU 2025

H₂エステル A : $C_{16}H_{18}O_6$ $U=8 = -\overset{\text{O}}{\parallel}{C}-O- \times 3 + \text{C}_6\text{H}_5 \times 1 + 1$

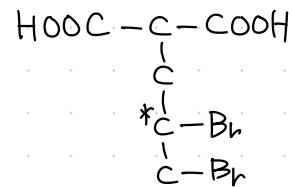
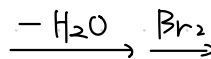
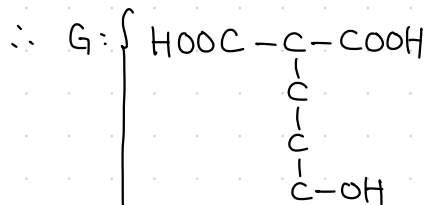
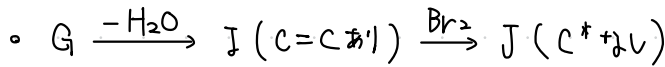
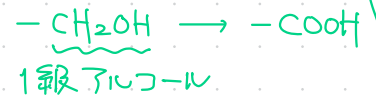
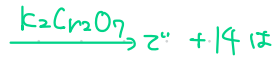
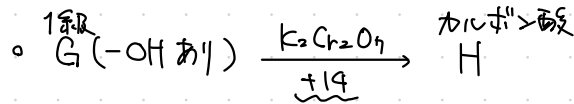
$\xrightarrow{KMnO_4}$ $2^\circ + 30$ は
 $-CH_3 \rightarrow -COOH$



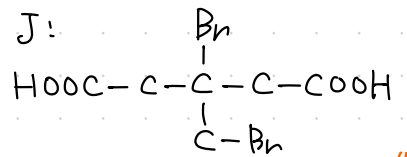
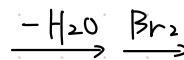
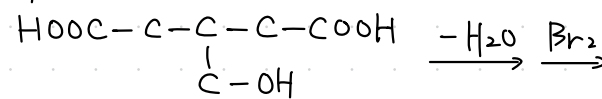


Bはトリエステル
だから,
Gはジカルボキ酸

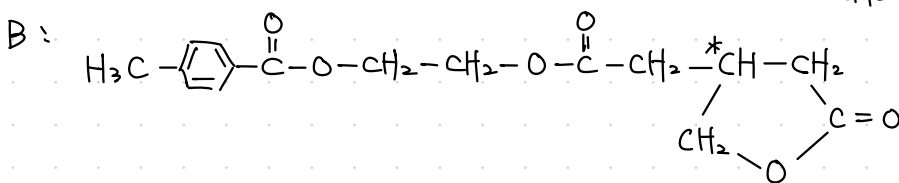
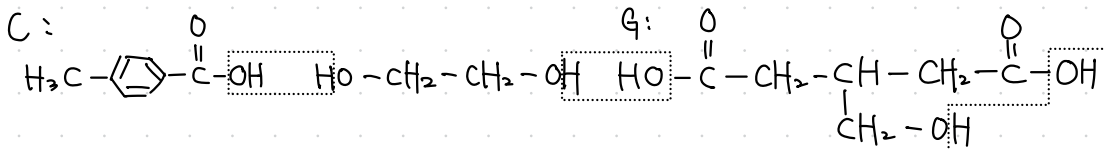
• 対称面あり
• $G = B^* + 3\text{H}_2\text{O} - (C + C_2\text{H}_6\text{O}_2) = C_6\text{H}_{10}\text{O}_5$ ($U=2$)



G:



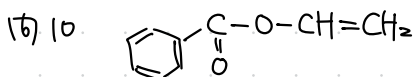
1676



1677

168 b. d

169 田舎



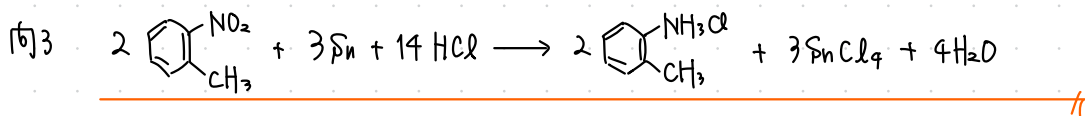
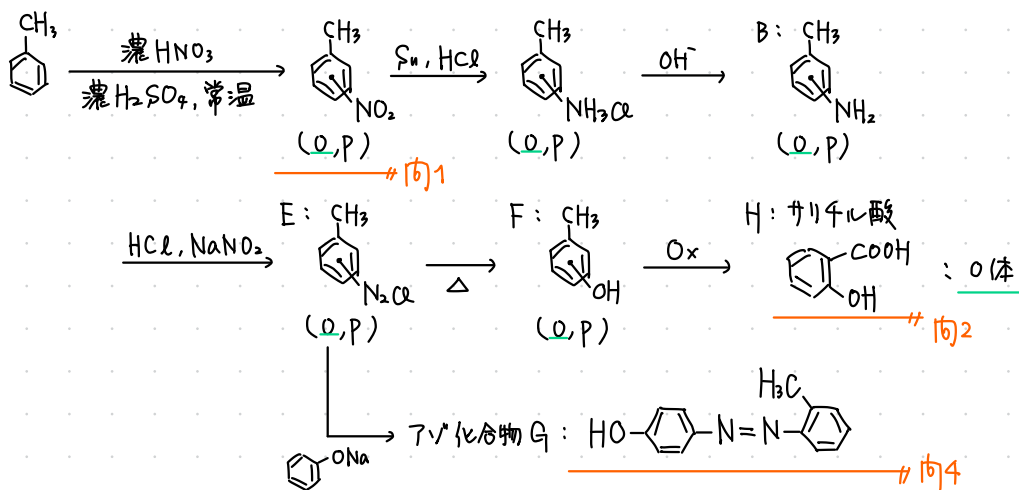
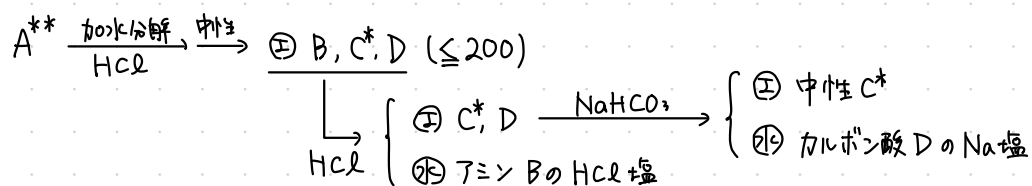
171 2.5 L $\left(\frac{222 \text{ g}}{148 \text{ g/mol}} \times n = 0.60 \text{ mol/L} \times V_L \right)$

172 (1) N: $\text{C}=\text{O} > \left(\xleftarrow{\text{H}^+} \text{M: PVA} \right)$

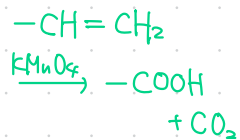
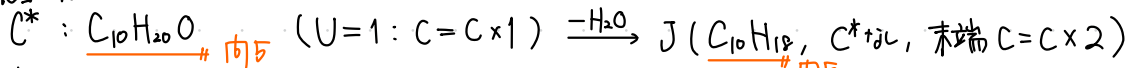
(2) $\frac{22 \text{ g}}{44 \text{ g/mol}} \times (44 + 6 \times 0.32)n \approx 23 \text{ g}$

2 THU 2024

A^{**} : $C_xH_yO_zN_w$, C^* 2つは 4級C原子ではない。



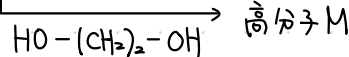
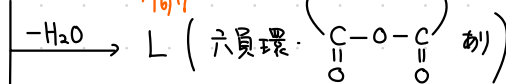
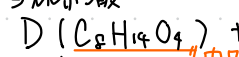
1級
アルコール



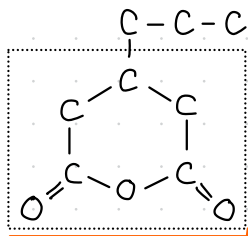
$\downarrow H_2$

K (C^*_{12} = 対称面あり)

$\xrightarrow{KMnO_4}$ ジカルボン酸

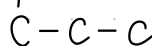
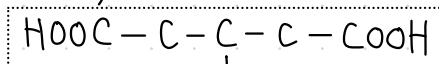


L:
(C_8)

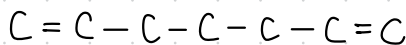


$\xleftarrow{\text{向8}}$

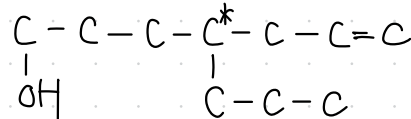
D (C_8)



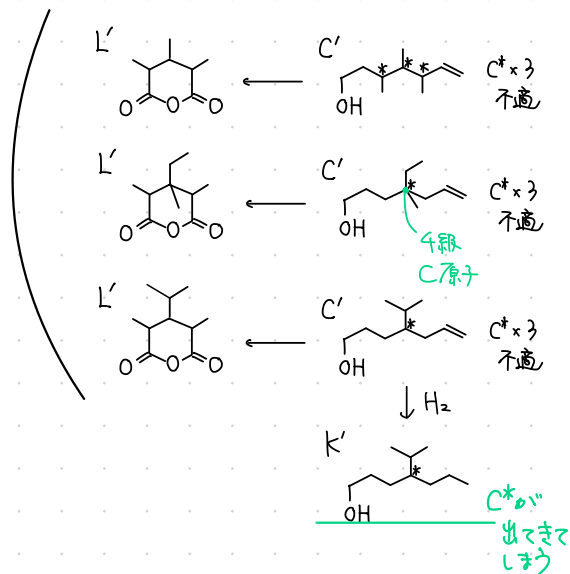
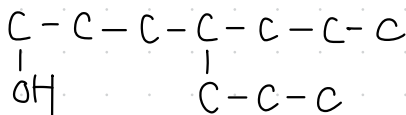
J (C_{10})



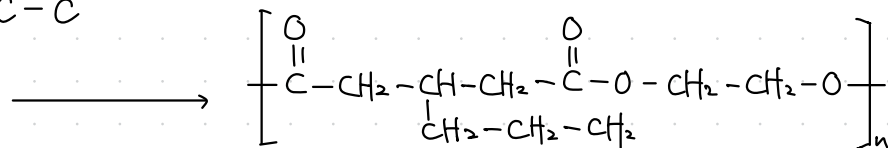
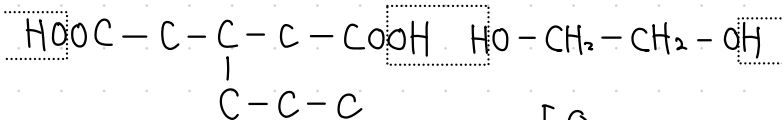
C (C_{10})



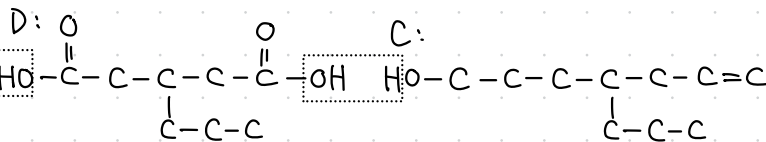
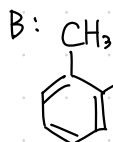
K (C^*_{12})



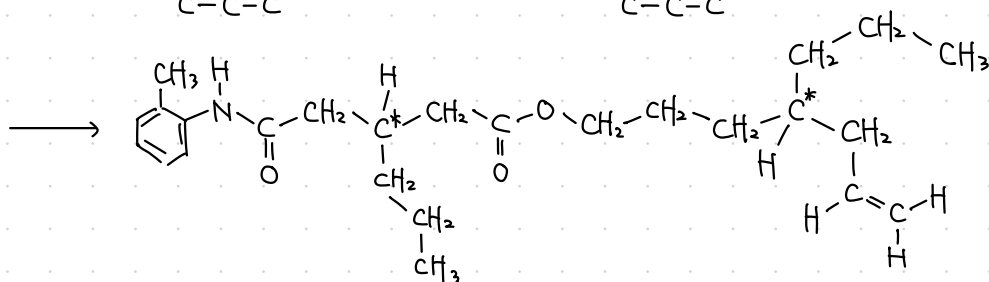
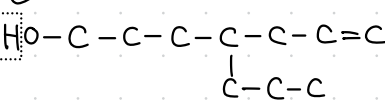
D:



$\xrightarrow{\text{向9}}$



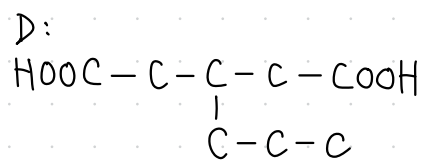
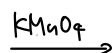
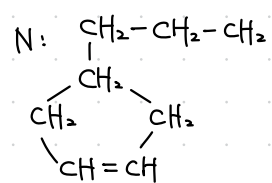
C:



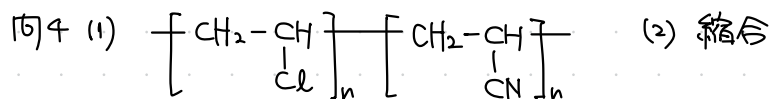
$\xrightarrow{\text{向12}}$

16) 10 2.4×10^2 g

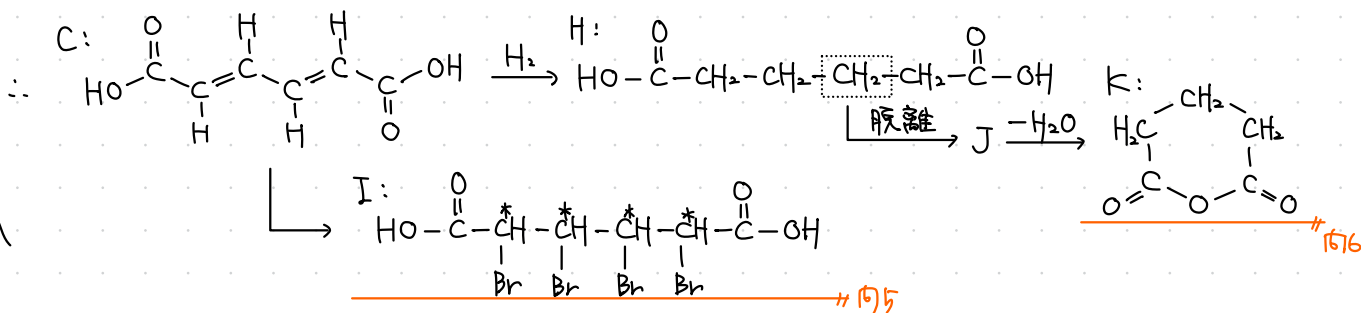
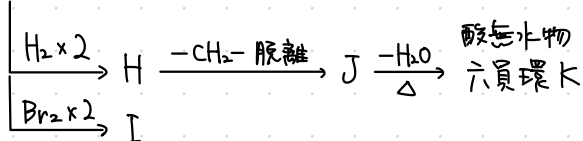
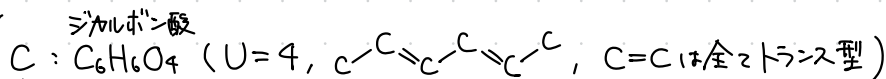
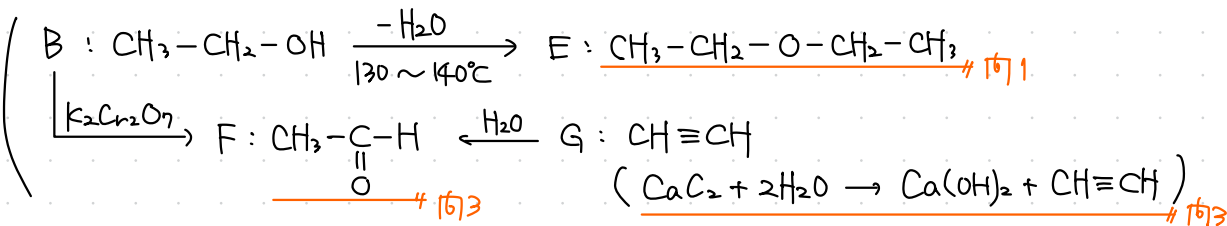
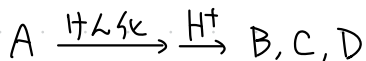
16) 11 C_8H_{14} $U=2$



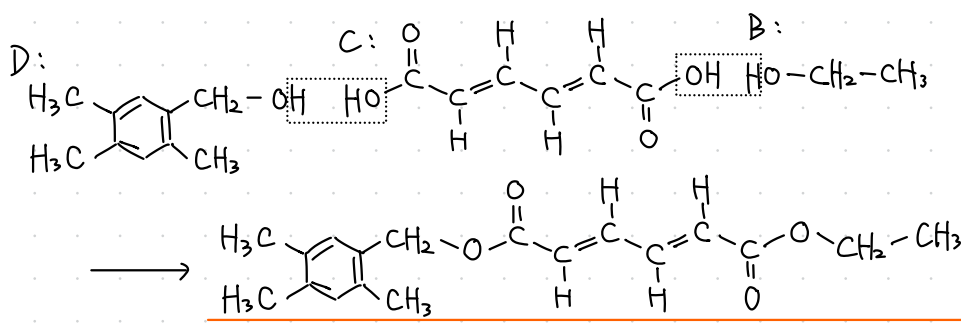
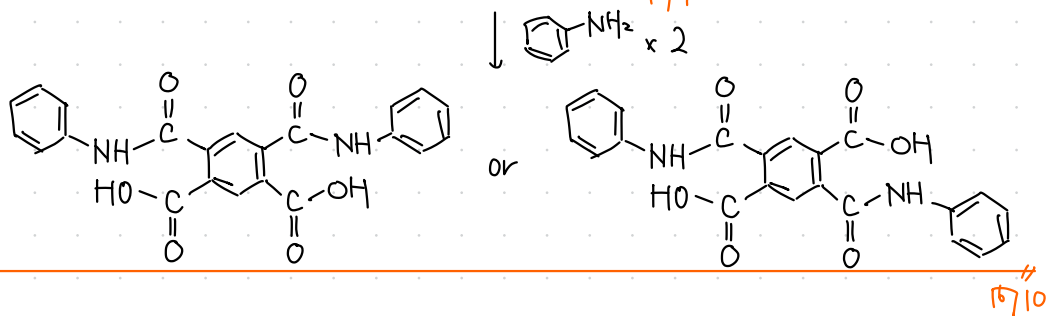
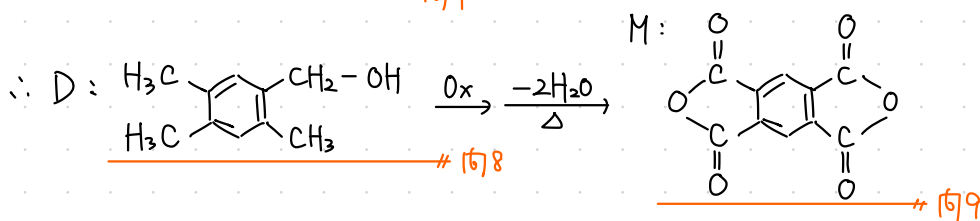
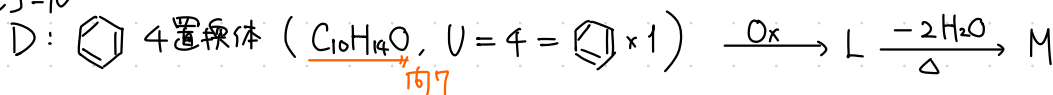
3 THU 2023



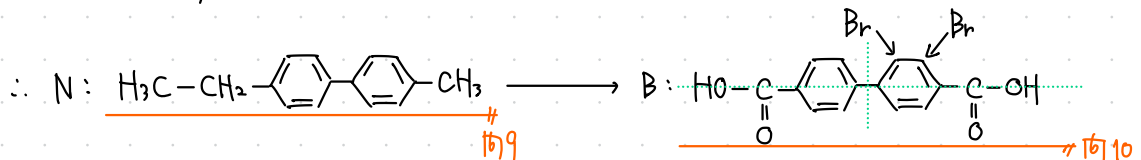
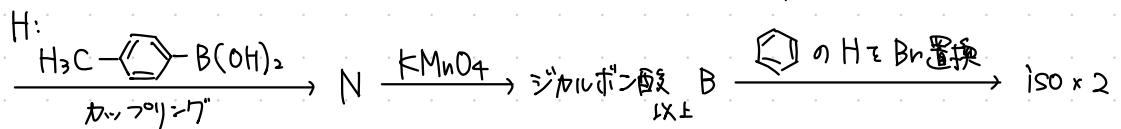
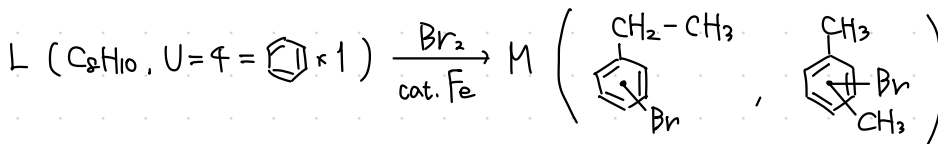
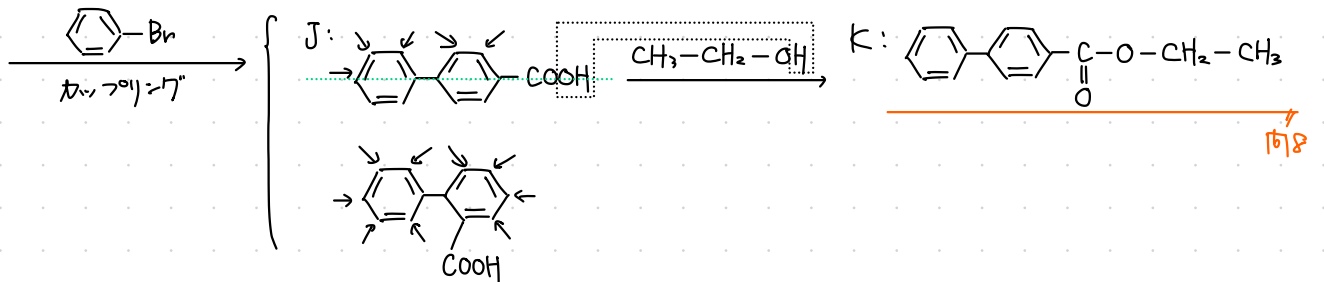
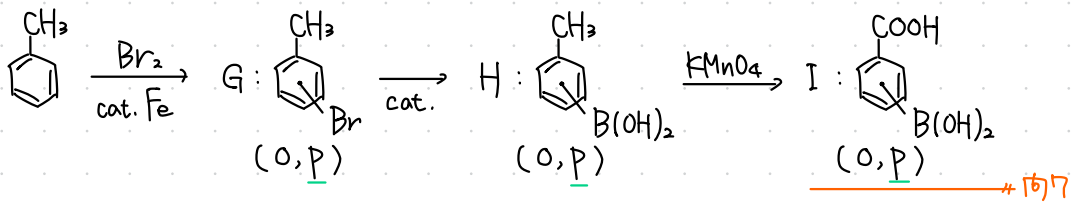
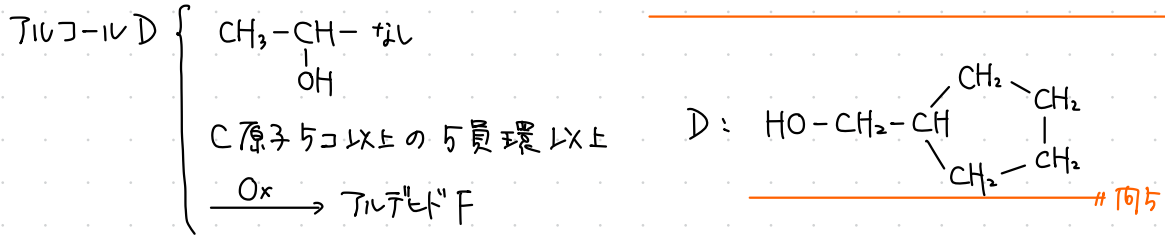
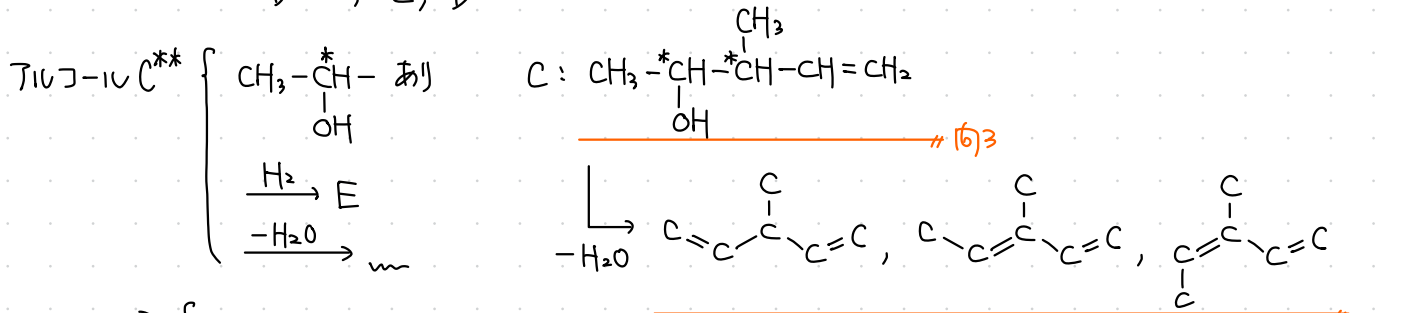
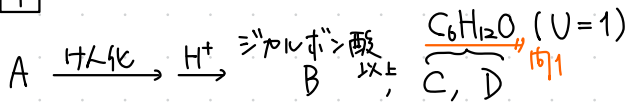
(3) $2.00 \times 10^2 : 1.00 \times 10^2$



第1級
アルコ-ル

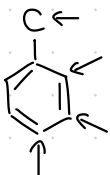


4



162 田舎

166

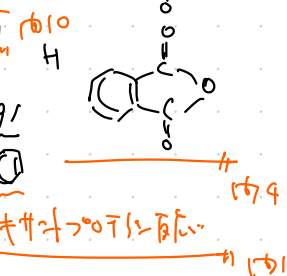
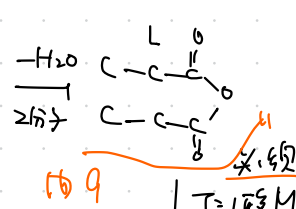
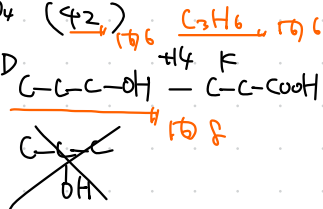
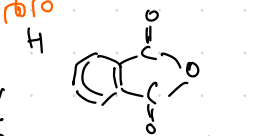
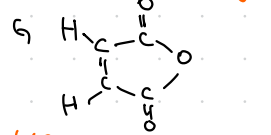
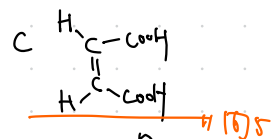
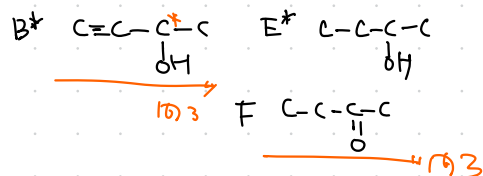


4

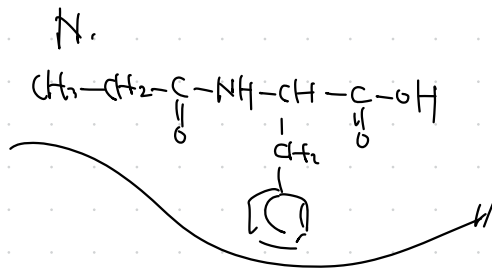
$\text{C}-\text{C}-\text{C}-\text{C}$
 $\uparrow \quad \uparrow$
 $(2) \quad (1)$

$\text{C}-\text{C}-\text{C}$
 $\uparrow \quad \uparrow$
 $(3) \quad (1)$

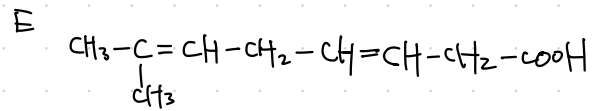
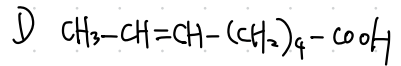
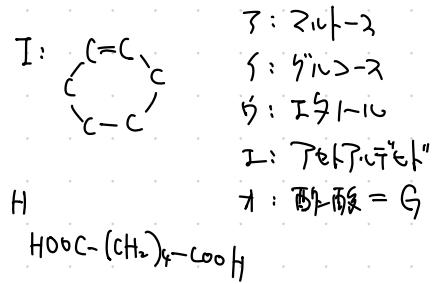
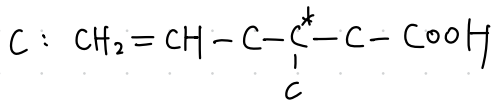
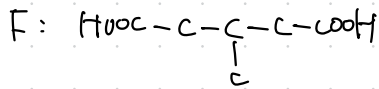
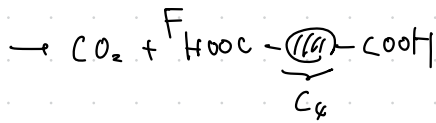
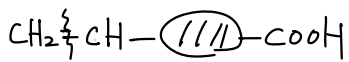
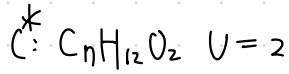
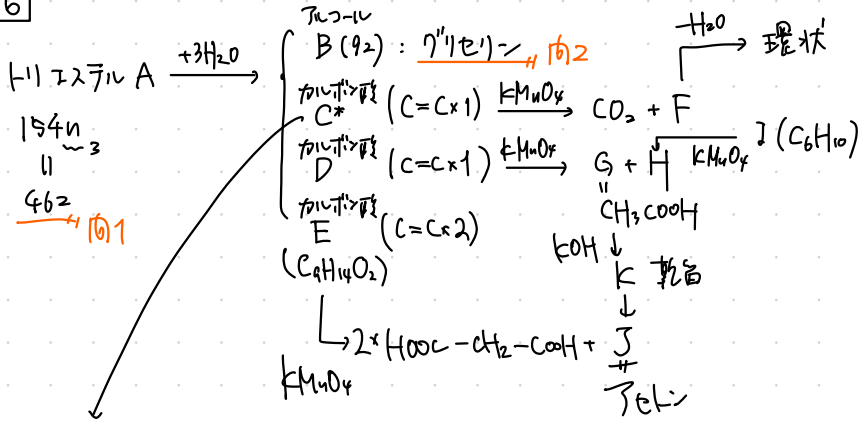
6



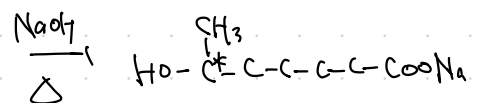
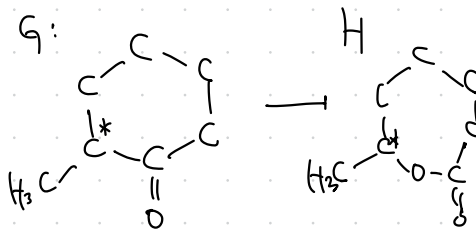
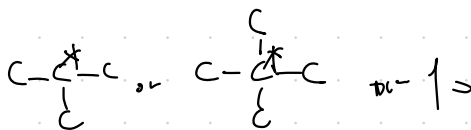
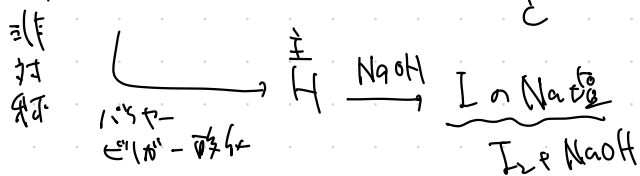
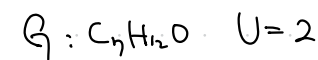
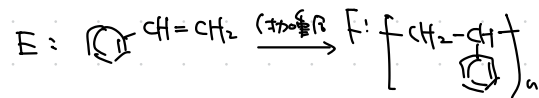
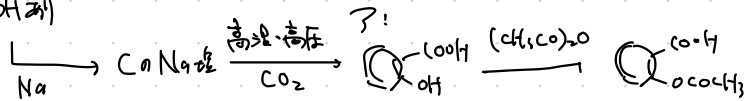
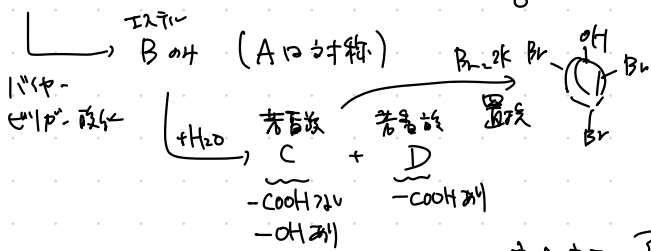
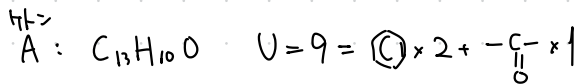
$N = 74 + 91$
 $-CH_2-$ 



6

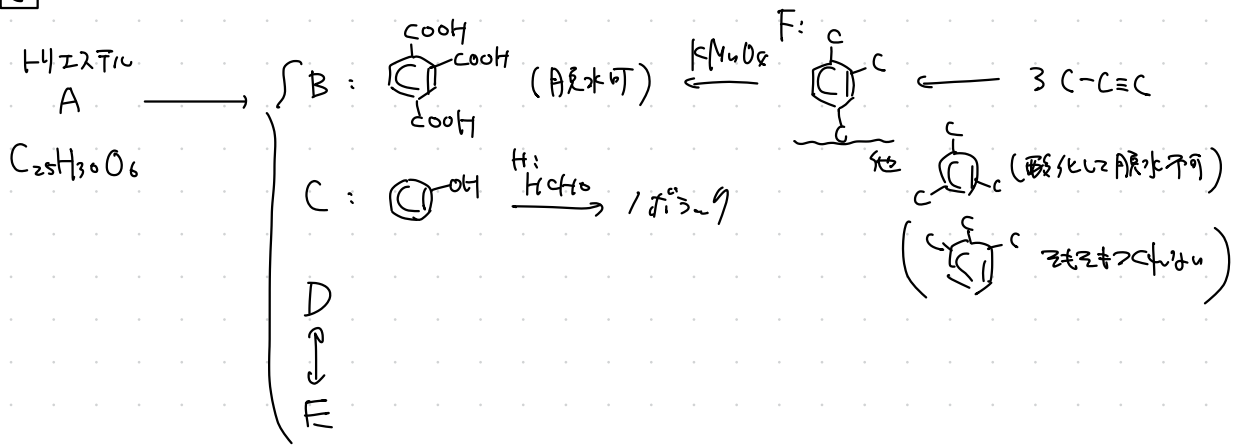


7

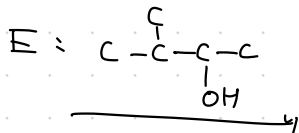
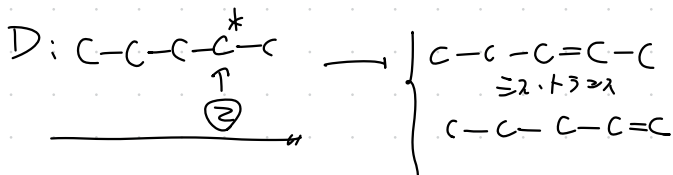


ア: e i: c り: h i: b

8

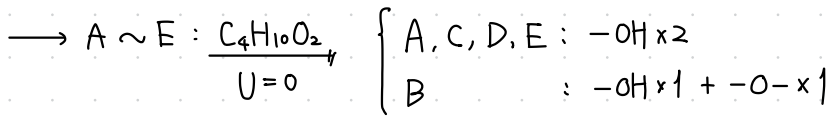


$$\text{D, E: } \frac{\text{C}_{25}\text{H}_{30}\text{O}_6 + 3\text{H}_2\text{O} - \text{C}_9\text{H}_6\text{O}_6 - \text{C}_6\text{H}_6\text{O}}{2} = \text{C}_5\text{H}_{12}\text{O}$$

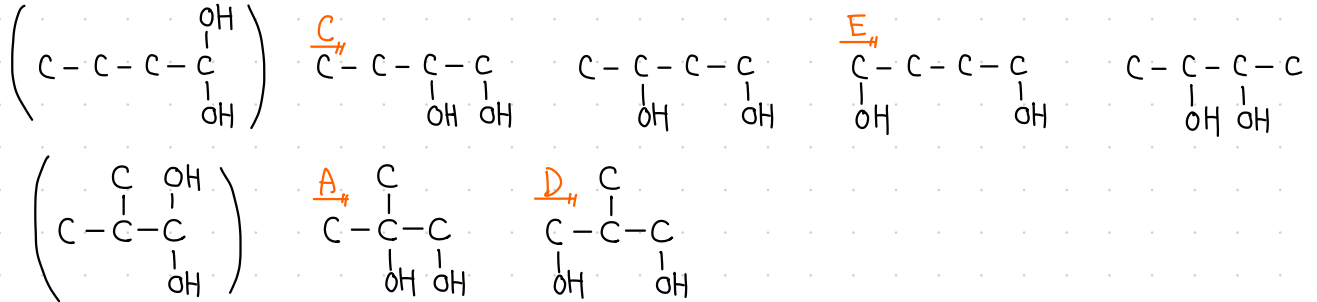


9

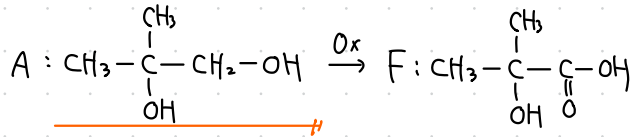
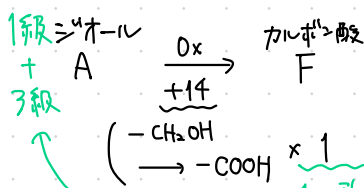
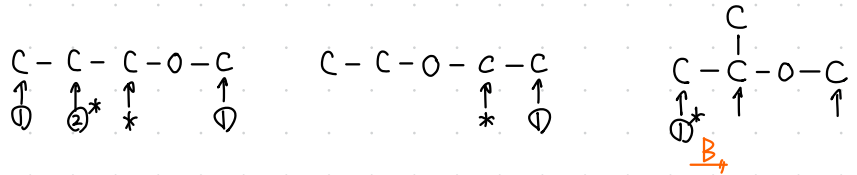
$$(C_2H_5O)_n \rightarrow \text{H数} \leq \frac{2 \cdot 2n + 2}{\text{最大H数}} \therefore n \leq 2 \quad \text{H数は偶数} \therefore n=2$$



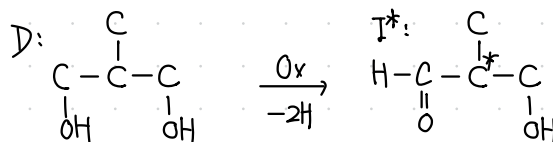
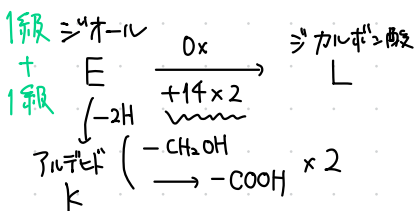
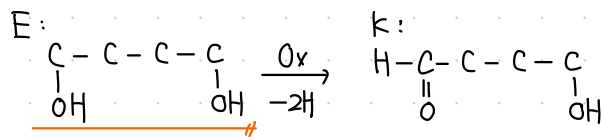
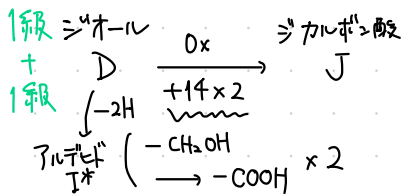
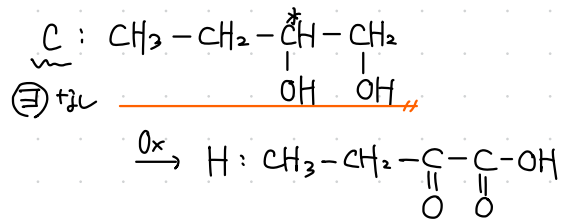
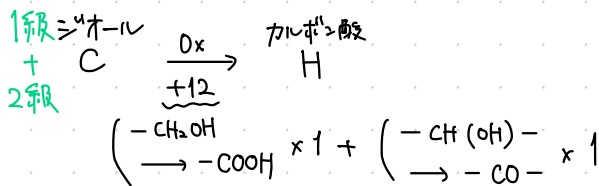
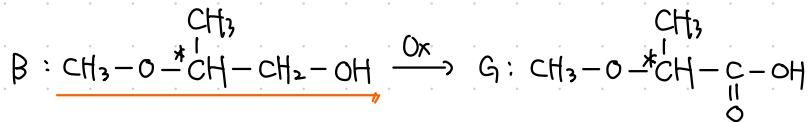
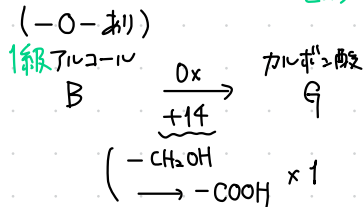
< A, C, D, E : -OH x 2 >

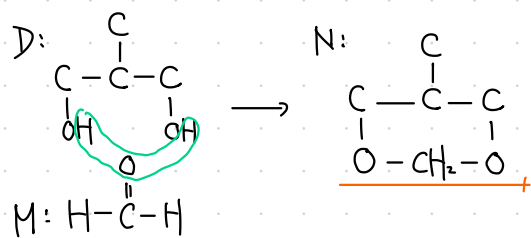


< B : -OH x 1 + -O- x 1 >



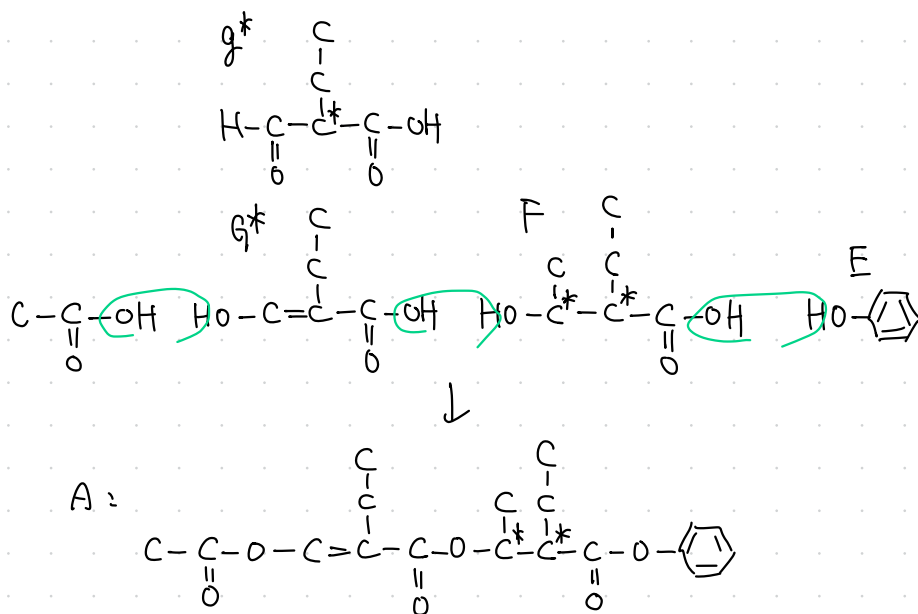
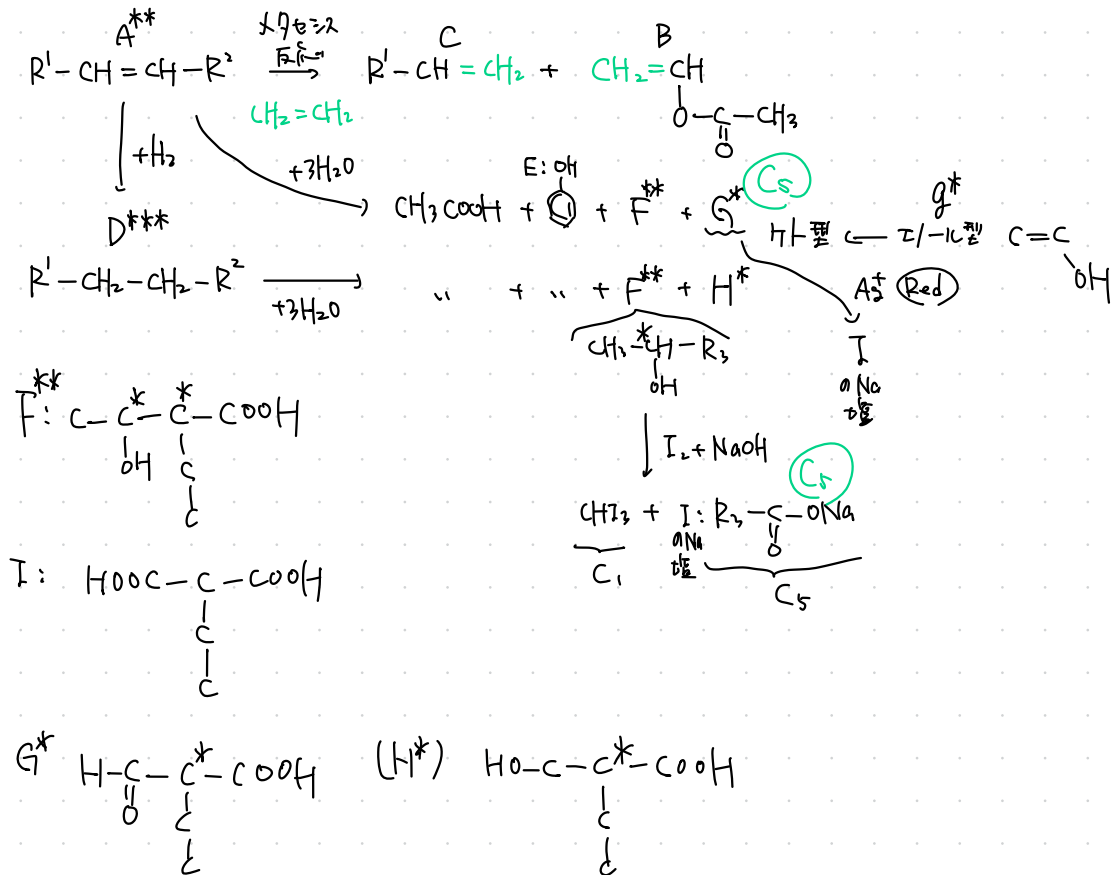
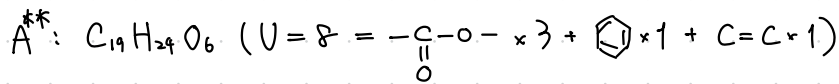
1か所酸化で済む
とこのことは、3級アルコール構造あり



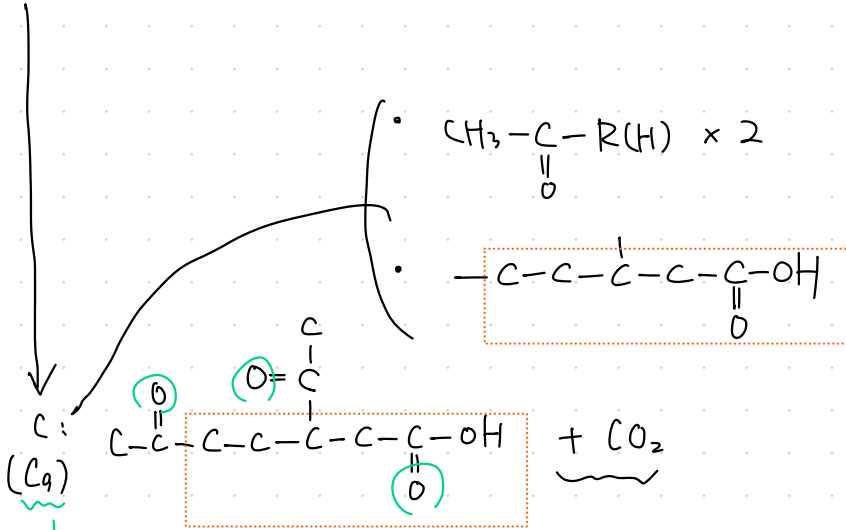
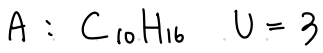


10 449

10



16



○ のうち 2ヶ所は 分子内 $\begin{cases} \text{C}=\text{C} \\ \text{環} \end{cases}$ $(U=2)$ + の2ヶ所 末端 $-\text{C}=\text{CH}_2$ $(U=1)$

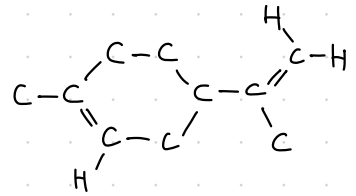
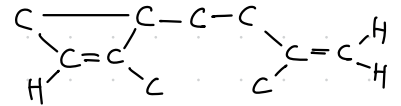
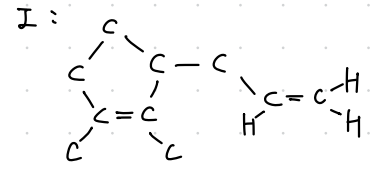


1: 二重結合 2つ

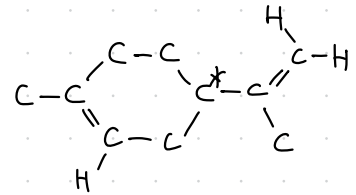
環 1つ

≡ 重結合 1つ

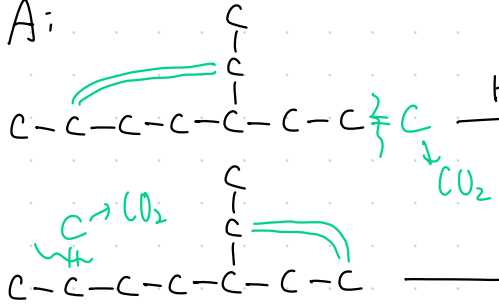
ウ: 3 4 3 3



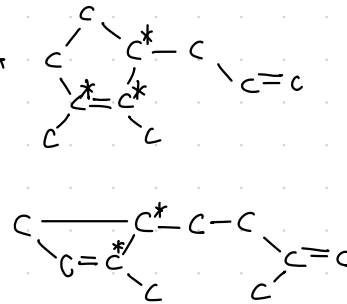
オ



A:



B



$\text{CO}_2 \leftarrow \text{C}_3$

